Fundamentals of Machine learning

M. Manasa
19241589
MLA0201

CONTINOUS ASSESSMENT (1)

Given data,

Prevalence of disease $C = 3\% = 0.03$

Probability of no disease $= 97\% = 0.97$

sensitivity $= 90\% = 0.90$

specificity $= 84\% = 0.84$.

false positive rate $= 1 - 0.84 = 0.16$

$P(T+) = P(T+|C)P(C) = P(T+|\bar{C})P(\bar{C})$

$= (0.90)(0.03) + (0.16)(0.97)$

$= 0.027 + 0.1552$.

$P(T+) = 0.1822 = 18.22\%$

43/50

Probability negative Test :-

$P(T-) = 1 - P(T+)$

$= 1 - 0.1822$

$\boxed{P(T-) = 0.8178 = 81.78\%}$

$$P(C|T+) = \frac{P(T+|C)P(C)}{P(T+)} = \frac{0.90 \times 0.03}{0.1822}$$

$$= \frac{0.027}{0.1822}$$

$P(C|T+) = 0.1482 = 14.82\%$.

Posterior probability

$$P(C|T-) = \frac{P(T-|C)P(C)}{P(T-)} = \frac{0.10 \times 0.09}{0.8178} = \frac{0.009}{0.8178}$$

$$P(C|T-) = 0.00367 = 0.367\%$$

For positive Test (T+):

$$R(\text{Treat}) = 2 \times P(C|T+)$$
$$= 2(1 - 0.1482)$$
$$= 1.7036$$

$$R(\text{Not}) = 50 \times P(L|T+)$$
$$= 50(0.1482)$$
$$= 7.4094$$

min visit = 1.7036

For Negative risk (T-)

$$R(\text{Treat}) = 2 \times P(L|T-)$$
$$= 2(1 - 0.00367)$$
$$= 1.9927$$

$$R(\text{Not}) = 50 \times (L|T-)$$
$$= 50(0.00367)$$
$$= 0.1834$$

min risk = 0.1834.

Overall Bayes risk

$$R = P(T+)R(T+) + P(T-)R(T-)$$

$$= (0.1822 \times 1.7036) + (0.8178 \times 0.1834)$$

$$= 0.4604$$

Probability after second Independent +ve test

$$P(T|T+, T+)$$

$$= \frac{0.1482 \times 0.90}{(0.1482 \times 0.90) + (0.8518 \times 0.16)} = 49.46\%$$

2) Given

Load X	Y = NO	Y = Yes
Low	0.30	0.02
medium	0.28	0.10
High	0.10	0.20

marginal probability of x:

$P(\text{low}) = 0.30 + 0.02 = 0.32$

$P(\text{medium}) = 0.28 + 0.10 = 0.38$

$P(\text{high}) = 0.10 + 0.20 = 0.30$

$P(X) = (0.32, 0.38, 0.30)$

$P(Y=) \ (0.68, 0.32)$

Entropy $H(X)$:

$$H(X) = -\Sigma p(x) \log_2 p(x)$$

$$\boxed{H(X) = 1.578 \text{ bites}}$$

Entropy $H(Y)$:

$$H(Y) = -\Sigma p(y) \log_2 p(y)$$

$$\boxed{H(Y) = 0.904 \text{ bites}}$$

Join entropy:

$$H(X,Y) = -\Sigma p(x,y) \log_2 p(x,y)$$

$$\boxed{H(X,Y) = 2.277 \text{ bites}}$$

Conditional entropy

$$H(Y|X) = H(X,Y) = H(X)$$

$$= 2.277 - 1.578$$

$$0.699 \text{ bites}.$$

given $Q = (0.5, 0.3, 0.2)$

$$H(P,Q) = -\Sigma p(x) \log_2 Q(x)$$

$$H(P,Q) = 1.677 \text{ bites}.$$

Fault probability

$$P(\text{Yes} | \text{High}) = \frac{0.20}{0.30}$$

$$\boxed{66.67\%}$$

∴ High network load has a higher probabilit

)(i) Temperature readings : 48, 52, 55, 47, 53, 51

$n = 6$

1. mean $= \bar{x} = \frac{48 + 52 + 55 + 47 + 53 + 51}{6}$

$\boxed{\bar{x} = 51}$

ML variance

$$\sigma^2 = \frac{\Sigma (x_i - \bar{x})^2}{n} = 46/6$$

$\boxed{\sigma^2 = 7.667}$

unbiased sample variance :

$$s^2 = \frac{46}{6-1}$$

$\boxed{s^2 = 9.2}$

probability Reading exceeds 55°C

$$z = \frac{55 - 7}{\sqrt{4.667}} \Rightarrow 1.445$$

$P(X > 55) = 7.43\%$

obability blw 46°C and 54°C

$$z_1 = \frac{46 - 51}{2.769} = -1.806$$

$$z_2 = \frac{54 - 51}{2.769} = 1.083$$

$P(46 < x < 54) = 0.8252 = \boxed{82.52\%}$

(ii) Given

$x = 10, 20, 30, 40, 50$

$y = 25, 38, 52, 61, 70$

$y = w_0 + w_1 x$

$w_1 = 1.31$

$w_0 = 11.7$

$\therefore \boxed{y = 11.7 + 1.31x}$

Sum of squared errors

$= \Sigma (y - \hat{y})^2$

$\boxed{SSE = 13.9}$

$y = 11.7 + 1.31(60) = \boxed{90.3ms}$

$\boxed{x = 500}$

$w_1 = \frac{1310}{1000 + 500} = 0.8733$

$w_0 = 51 - (0.8733 \times 30) = \boxed{24.80}$

$y = 24.8 + 0.8733 \quad (60)$

$\boxed{y = 77.2ms}$

Given data

Total emails = 100

spam = 40

legitimate = 60.

word	spam	legitimate
offer	28	6
free	24	3
meeting	4	33

a) Calculate class priors and conditional probabilities.

Total emails = 100.

spam = 40 = 0.4

legitimate = 60 = 0.60.

Spam (word | offer) = 0.4 × 0.60 =

spam (word | free) = 24 × 3 = 72

spam (word | meeting) = 4 × 33 = 132.

$$P(\text{spam} = \frac{40}{100} = 0.4$$

legitimate emails (N (legit) = 60

= 0.6

Conditional probability formula:

$$P(W = 1|C) = \frac{N_{WC} + 1}{N_C + 2}$$

b) classification of new email.

The new email is $x = [offer = 1, free = 1, meeting = 0]$.

→ $score(spam) = P(spam) \cdot P(offer = 1 | spam)$

$= 0.4 \times 0.6905 = 0.5952(1 - 0.1190)$

$= 0.4 \cdot 0.6905 = 0.5952 \cdot 0.8810$

$= 0.144828.$

$score|legit = P(legit) \cdot P(offer = 1 | legit) \cdot P(free = 1 | legit)$

$(1 - P(meeting = 1 | legit))$

$= 0.6 \times 0.1129 \times 0.0645 \times (1 - 0.5484) = 0.001974.$

posterior probability spam

$$P(spam | x) = \frac{score(spam)}{score(spam) + score(legit)} = \frac{0.144828}{0.144828 + 0.0019}$$

$= 0.9866.$

$$= \frac{score(not\ spam)}{score(not\ spam) + score(legit)} = \frac{0.12834}{0.367 \quad 0.28}$$

Since $P(spam | x) = 98.66\% > 50\%.$

the email is classified as spam.

cost - sensitive threshold determination

using Baye's theorem.

$$P(\text{legit} \mid X) \cdot \lambda_{FP} < P(\text{spam} \mid X) \cdot \lambda_{FN}$$

$$(1 - P(\text{spam} \mid X)) \cdot 5 < P(\text{spam} \mid X) \cdot 1 \Rightarrow 5 - 5P < P$$

$$\Rightarrow 6P > 5$$

$$\text{Threshold} = \frac{5}{5+1} = \frac{5}{6} = 0.8333,$$

decision for new email = since $P(\text{spam} \mid X) =$ 0.9866 is greater than cost - sensitive threshold of 0.8333. The decision is remain to classify the email as spam.

d) Evaluation and Discussion

TP = 45, FN = 15, FP = 10, TN = 130, Total = 200

Accuracy :-

$$\frac{TP + TN}{\text{Total}} = \frac{45 + 130}{200} = 0.875$$

Precision.

$$\frac{TP}{TP + FP} = \frac{45}{45 + 10} = 0.818$$

Recall

Recall:

$$\frac{TP}{TP+FN} = \frac{45}{45+15} = 0.750$$

Fiscore:-

$$2 \cdot \frac{\text{precision} \cdot \text{Recall}}{\text{precision} + \text{recall}} = 2 \cdot \left[\frac{0.614}{1.568}\right] = 2 \times [0.392]$$

$$= 0.784.$$

Calculated Accuracy, precision, recall and Fiscore.

$$\text{Accuracy} = \frac{TP+TN}{\text{Total}} = \frac{45+130}{200} = 0.875$$

$$\text{Precision} = \frac{TP}{TP+fp} = \frac{45}{45+10} = 0.818$$

$$\text{Recall} \geq \frac{TP}{TP+FN} = \frac{45}{45+15} = 0.750$$

$$\text{Fiscore} = 2 \cdot \frac{\text{precision} \cdot \text{Recall}}{\text{precision} + \text{recall}.} = 2 \cdot \left[\frac{0.614}{1.568}\right] = 0.78$$

Accuracy = 0.875, precision = 0.818, Recall = 0

Fiscore = 0.784

5) Given

$x_1 = 0.5, x_2 = 1.0$

$t = 1$

$\eta = 0.5$

$E = \frac{1}{2}(t-y)^2$

$f(z) = \frac{1}{1+e^{-z}}$

$f'(z) = f(z)(1 - f(z))$

Calculate h_1:

$net_1 = (0.5 \times 0.4) + (1 \times -0.2) + 0.1$

$net_1 = 0.1$

$h_1 = \frac{1}{1+e^{-0.1}}$

$\boxed{h_1 = 0.5250}$

$h_2 = (0.5)(0.3) + (1 \times 0.5) - 0.1$

$= 0.55$

$\boxed{h_2 = 0.6341}$

Calculate output and error.

$net_y = (0.5250 \times 0.6) + (0.6341 \times -0.4) + 0.2$

$net_y = 0.2613$

$$y = \frac{1}{1+e^{-0.2613}}$$

$$y = 0.5650$$

error:

$$E = \frac{1}{2}(1-0.5650)^2$$

$$E = 0.09463$$

output $\Delta S_y = (t-y)y(1-y)$

$$S_y = 0.10692$$

unit 1:-

$$S_{h1} = h_1(1-h_1)v_1 S_y$$

$$S_h = 0.01600$$

$$S_{h2} = h_2(1-h_2)v_2 S_y$$

$$S_{h2} = -0.00992$$

weights & bias

parameter	ordvalue	newvalue
w_{11}	0.4	0.4000
w_{21}	−0.2	−0.1920
b_1	0.1	0.1080
b_{12}	0.3	0.2975
w_{22}	0.5	0.4950
b_2	−0.1	−0.1050
v_1	0.6	0.6281
v_2	−0.4	−0.3661
b_3	0.2	0.2885

location

temper

h

forward pass

$h = 0.5295$

$h_2 = 0.6315$

$E_{new} = \frac{1}{2}(1 - 0.5878)^2$

$E_{new} = 0.84986$

Error reduction

$\% = \frac{\text{old error} - \text{new error}}{\text{old error}} \times 100$

$= \frac{0.09463 - 0.08496}{0.09463} \times 100$

10.21%

$=$